

Demographic Insights Document: Student Depression Dataset Analysis

1. Introduction

This document presents demographic insights from the Student Depression Dataset as part of Task 10 (T10) in our project plan. The analysis examines various demographic factors and their relationship with depression rates among students, providing a foundation for targeted interventions and support strategies.

2. Gender Distribution and Depression Rates

2.1 Overall Gender Distribution

- **Male:** 15,547 students (55.72% of sample)
- **Female:** 12,354 students (44.28% of sample)

2.2 Depression Rates by Gender

- **Male:** 58.63% depression rate (9,115 depressed out of 15,547)
- **Female:** 58.45% depression rate (7,221 depressed out of 12,354)

2.3 Insight

The depression rates are nearly identical between male and female students, suggesting that gender alone may not be a significant predictor of depression in this population.

3. Age Distribution and Depression Rates

3.1 Age Group Distribution

Age Group	Count	Percentage
18-20	5,384	19.30%
21-25	8,573	30.73%
26-30	7,845	28.12%
31-35	6,060	21.72%
36+	39	0.14%

3.2 Depression Rates by Age Group

Age Group	Depression Rate
18-20	72.34%
21-25	64.29%
26-30	56.53%
31-35	40.97%
36+	28.21%

3.3 Insight

There is a clear negative correlation between age and depression rates. Younger students (18-20) have significantly higher depression rates (72.34%) compared to older students. This suggests that younger students may be more vulnerable to academic and social pressures, or may have less developed coping mechanisms.

4. Geographic Distribution and Depression Rates

4.1 Top 10 Cities by Student Population

- 1. Kalyan: 1,570 students (5.63%)
- 2. Srinagar: 1,372 students (4.92%)
- 3. Hyderabad: 1,340 students (4.80%)
- 4. Vasai-Virar: 1,290 students (4.62%)
- 5. Lucknow: 1,155 students (4.14%)
- 6. Thane: 1,139 students (4.08%)
- 7. Ludhiana: 1,111 students (3.98%)
- 8. Agra: 1,094 students (3.92%)
- 9. Surat: 1,078 students (3.86%)
- 10. Kolkata: 1,066 students (3.82%)

4.2 Depression Rates by Region

Region	Student Count	Depression Rate
East	2,073	61.99%
South	3,961	61.70%
West	9,752	58.74%
Central	3,264	57.41%
North	8,825	56.54%

4.3 Cities with Highest Depression Rates (among top 10)

- 1. Hyderabad: 66.94%
- 2. Kolkata: 60.88%
- 3. Thane: 59.09%
- 4. Kalyan: 59.49%

4.4 Insight

There are notable regional differences in depression rates, with Eastern and Southern regions showing higher rates compared to Northern regions. This may reflect differences in academic environments, cultural factors, or mental health support systems across regions.

5. Educational Background and Depression Rates

5.1 Depression Rates by Education Level

Education Level	Student Count	Depression Rate
Undergraduate	18,048	61.19%
Doctorate	522	54.79%
Postgraduate	8,625	53.67%
Other	706	53.40%

5.2 Top Degree Programs by Depression Rate

- 1. Class 12: 70.77% (4,303 depressed out of 6,080)
- 2. B.Arch: 58.93% (871 depressed out of 1,478)
- 3. B.Tech: 56.86% (655 depressed out of 1,152)
- 4. B.Com: 56.64% (853 depressed out of 1,506)
- 5. BCA: 57.15% (819 depressed out of 1,433)

5.3 Insight

Undergraduate students, particularly those in Class 12 (pre-university), have significantly higher depression rates compared to postgraduate and doctoral students. This suggests that the transition to university education and early academic pressure may be particularly challenging periods for mental health.

6. Lifestyle Factors by Demographics

6.1 Sleep Duration and Depression Rates

Sleep Duration	Student Count	Depression Rate
Less than 5 hours	8,310	64.51%
5-6 hours	6,183	56.88%
7-8 hours	7,346	59.50%
More than 8 hours	6,044	50.93%

6.2 Dietary Habits and Depression Rates

Dietary Habits	Student Count	Depression Rate
Unhealthy	10,317	70.73%
Moderate	9,921	56.02%
Healthy	7,651	45.39%

6.3 Insight

Sleep duration and dietary habits show strong associations with depression rates. Students with less than 5 hours of sleep have higher depression rates (64.51%) compared to those with more than 8 hours (50.93%). Similarly, students with unhealthy dietary habits have significantly higher depression rates (70.73%) compared to those with healthy habits (45.39%). These lifestyle factors appear to be important correlates of mental health.

7. Gender Differences in Key Academic and Stress Factors

Factor	Male Average	Female Average
Academic Pressure	3.11/5	3.18/5
Study Satisfaction	2.92/5	2.97/5
Work/Study Hours	7.20 hrs	7.10 hrs
Financial Stress	3.13/5	3.15/5
CGPA	7.70/10	7.60/10

7.1 Insight

Despite similar depression rates, there are small differences in academic and stress factors between genders. Female students report slightly higher academic pressure and financial stress, but also slightly higher study satisfaction. Male students report slightly higher CGPA and work/study hours. However, these differences are minimal and may not be practically significant.

8. Suicidal Thoughts Analysis

8.1 Overall Distribution

- **Students reporting suicidal thoughts:** 17,656 (63.28%)
- **Students not reporting suicidal thoughts:** 10,245 (36.72%)

8.2 Depression Rates by Suicidal Thoughts

- **Among students with suicidal thoughts:** 79.05% depression rate
- **Among students without suicidal thoughts:** 23.22% depression rate

8.3 Insight

The strong correlation between suicidal thoughts and depression (79.05% of students with suicidal thoughts are classified as depressed) underscores the critical nature of mental health support on campus. The presence of suicidal thoughts appears to be one of the strongest indicators of depression in this dataset.

9. Key Demographic Insights

1. **Age is a significant factor:** Younger students (18-20) have significantly higher depression rates, suggesting they may need targeted support.
2. **Education level matters:** Undergraduate students, particularly those in Class 12, show higher depression rates than postgraduate students.

3. **Lifestyle factors are important:** Sleep duration and dietary habits show strong correlations with depression rates, with unhealthy habits associated with higher depression rates.
4. **Regional differences exist:** Eastern and Southern regions show higher depression rates compared to Northern regions.
5. **Gender is not a significant predictor:** Despite small differences in academic factors, depression rates are nearly identical between male and female students.
6. **Suicidal thoughts are strongly linked to depression:** Students reporting suicidal thoughts have much higher depression rates, highlighting the need for urgent mental health interventions.

10. Recommendations for Further Analysis

1. Investigate the factors contributing to higher depression rates among younger students (18-20).
2. Explore the intersection of demographic factors (e.g., age, education level, and lifestyle factors) to identify high-risk student profiles.
3. Analyze regional differences in more detail to understand contextual factors that may influence depression rates.
4. Examine the relationship between academic pressure, CGPA, and depression rates across different degree programs.
5. Investigate protective factors that might explain why some students maintain good mental health despite academic pressures.
6. Design targeted interventions for high-risk demographic groups, particularly focusing on lifestyle factors (sleep, diet) that could be modified.